✅ 1. Gumbel model (μ, σ)

$$f(x) = \frac{1}{\sigma} \exp\left(-\frac{x-\mu}{\sigma}\right) \exp\left(-e^{-\frac{x-\mu}{\sigma}}\right)$$

$$F(x) = \exp\left(-e^{-\frac{x-\mu}{\sigma}}\right)$$

Thus:

$$\log f(x) = -\log\sigma - \frac{x-\mu}{\sigma} - e^{-\frac{x-\mu}{\sigma}}, \quad \log F(x) = -e^{-\frac{x-\mu}{\sigma}}.$$

✅ 2. Log-likelihood

Starting from your likelihood, taking logs:

$$\boxed{\begin{aligned}\ell(\gamma,\mu,\sigma) = &\left(\sum_{n=1}^{N_T} l_n\right)\log\gamma \\ &+ \sum_{n=1}^{N_T}\log f(x_{l_n}) \\ &+ \sum_{n=1}^{N_T}(\gamma^{l_n}-1)\log F(x_{l_n}) \\ &+ \sum_{n=1}^{N_T-1}\frac{\gamma^{l_n+1}-\gamma^{l_{n+1}}}{1-\gamma}\log F(x_{l_n}) \\ &+ \mathbf{1}_{l_{N_T}<T}\frac{\gamma^{l_{N_T}+1}-\gamma^{T+1}}{1-\gamma}\log F(x_{l_{N_T}})\end{aligned}}$$

✅ 3. Substitute Gumbel

Define:

$$u_n = \frac{x_{l_n}-\mu}{\sigma}, \quad e_n = \exp(-u_n)$$

Then:

$$\boxed{\ell(\theta) = \left(\sum_{n=1}^{N_T} L_n\right)\log\gamma - N_T\log\sigma - \sum_{n=1}^{N_T} u_n - \sum_{n=1}^{N_T} w_n e_n,}$$

The weights w_n depend only on γ:

$$w_n = 1 + (\gamma^{L_n} - 1) + \frac{\gamma^{L_n+1} - \gamma^{L_{n+1}}}{1-\gamma}, \qquad n < N_T,$$

and

$$w_{N_T} = 1 + (\gamma^{L_{N_T}} - 1) + \frac{\gamma^{L_{N_T}+1} - \gamma^{T+1}}{1-\gamma}$$

if $L_{N_T} < T$, otherwise

$$w_{N_T} = \gamma^{L_{N_T}}.$$

✅ 5. Interpretation

This Gumbel version has a very nice structure:

$$\ell = \underbrace{\sum l_n \log \gamma}_{\text{trend}} - \underbrace{N_T \log \sigma}_{\text{scale}} - \underbrace{\sum u_n}_{\text{linear part}} - \underbrace{\sum w_n e_n}_{\text{extreme tail part}}$$

✅ Compared to Fréchet:

- exponential e^{-u_n} replaces power $(Ax)^{-\alpha}$
- numerically more stable
- easier derivatives

✅ 2. First-order derivatives

🔷 w.r.t. μ

$$\boxed{\frac{\partial \ell}{\partial \mu} = \frac{N_T}{\sigma} - \frac{1}{\sigma} \sum w_n e_n}$$

🔷 w.r.t. σ

$$\boxed{\frac{\partial \ell}{\partial \sigma} = -\frac{N_T}{\sigma} + \frac{1}{\sigma} \sum_n u_n - \frac{1}{\sigma} \sum_n w_n e_n u_n}$$

🔷 w.r.t. γ

Let $S = \sum l_n$

$$\boxed{\frac{\partial \ell}{\partial \gamma} = \frac{S}{\gamma} - \sum e_n \frac{\partial w_n}{\partial \gamma}}$$

with:

$$\frac{\partial}{\partial \gamma} \gamma^{l_n} = l_n \gamma^{l_n - 1}$$

Define

$$S_L = \sum_{n=1}^{N_T} L_n.$$

Then

$$\frac{\partial \ell}{\partial \gamma} = \frac{S_L}{\gamma} - \sum_n e_n \frac{\partial w_n}{\partial \gamma}.$$

Thus the problem reduces to $\partial w_n / \partial \gamma$.

For $n < N_T$,

$$w_n = \gamma^{L_n} + \frac{\gamma^{L_n+1} - \gamma^{L_{n+1}}}{1-\gamma}.$$

Hence

$$\frac{\partial w_n}{\partial \gamma} = L_n \gamma^{L_n - 1} + \frac{\left((L_n+1)\gamma^{L_n} - L_{n+1}\gamma^{L_{n+1}-1}\right)(1-\gamma) + \gamma^{L_n+1} - \gamma^{L_{n+1}}}{(1-\gamma)^2}.$$

For the last record,

$$\frac{\partial w_{N_T}}{\partial \gamma} = L_{N_T} \gamma^{L_{N_T} - 1} + \frac{\left((L_{N_T}+1)\gamma^{L_{N_T}} - (T+1)\gamma^{T}\right)(1-\gamma) + \gamma^{L_{N_T}+1} - \gamma^{T+1}}{(1-\gamma)^2}$$

when $L_{N_T} < T$.

Therefore

$$\boxed{\frac{\partial \ell}{\partial \gamma} = \frac{S_L}{\gamma} - \sum_n e_n w_n'}$$

with $w_n' = \partial w_n / \partial \gamma$.

✅ 3. Second-order derivatives

🔷 $\mu\mu$

$$\boxed{\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{1}{\sigma^2} \sum w_n e_n}$$

🔷 $\sigma\sigma$

$$\boxed{\begin{aligned}\frac{\partial^2 \ell}{\partial \sigma^2} = \frac{N_T}{\sigma^2} - \frac{2}{\sigma^3} \sum (x_{l_n} - \mu) \\ + \frac{2}{\sigma^3} \sum w_n (x_{l_n} - \mu) e_n \\ + \frac{1}{\sigma^4} \sum w_n (x_{l_n} - \mu)^2 e_n\end{aligned}}$$

$$\boxed{\frac{\partial^2 \ell}{\partial \sigma^2} = \frac{N_T - 2\sum_n u_n + \sum_n w_n e_n u_n (2 - u_n)}{\sigma^2}}$$

🔷 Terms involving γ

These include:

$$\frac{\partial^2 \ell}{\partial \gamma^2}, \frac{\partial^2 \ell}{\partial \gamma \partial \mu}, \frac{\partial^2 \ell}{\partial \gamma \partial \sigma}$$

$\gamma\gamma$

$$\boxed{\frac{\partial^2 \ell}{\partial \gamma^2} = -\frac{S_L}{\gamma^2} - \sum_n e_n w_n''}$$

where $w_n'' = \partial^2 w_n / \partial \gamma^2$.

For $n < N_T$,

$$w_n'' = L_n(L_n - 1)\gamma^{L_n - 2} + \frac{A_n'(1-\gamma)^2 + 2A_n(1-\gamma)}{(1-\gamma)^4},$$

with

$$A_n = \left((L_n + 1)\gamma^{L_n} - L_{n+1}\gamma^{L_{n+1}-1}\right)(1-\gamma) + \gamma^{L_n+1} - \gamma^{L_{n+1}}$$

and

$$A_n' = \left((L_n+1)L_n\gamma^{L_n-1} - L_{n+1}(L_{n+1}-1)\gamma^{L_{n+1}-2}\right)(1-\gamma) - \left((L_n+1)\gamma^{L_n} - L_{n+1}\gamma^{L_{n+1}-1}\right) +$$

The last observation follows by replacing L_{n+1} with $T + 1$.

3 Cross Derivatives

🔷 $\mu\sigma$

$$\boxed{\frac{\partial^2 \ell}{\partial \mu \partial \sigma} = -\frac{N_T}{\sigma^2} + \frac{1}{\sigma^2}\sum w_n e_n - \frac{1}{\sigma^3}\sum w_n (x_{l_n} - \mu) e_n}$$

$$\boxed{\frac{\partial^2 \ell}{\partial \mu \partial \sigma} = -\frac{N_T}{\sigma^2} + \frac{1}{\sigma^2}\sum_n w_n e_n (1 - u_n)}$$

$\mu\gamma$

Since only w_n depends on γ,

$$\boxed{\frac{\partial^2 \ell}{\partial\mu\partial\gamma} = -\frac{1}{\sigma}\sum_n e_n w_n'}$$

$\sigma\gamma$

$$\boxed{\frac{\partial^2 \ell}{\partial\sigma\partial\gamma} = -\frac{1}{\sigma}\sum_n e_n u_n w_n'}$$